CLASS PROJECT · NOVEMBER 2026

Contract Clause Review

Finds the clause, shows the exact words, and flags what it could not find.

Assignment | Change of control | Termination for convenience | Renewal | Governing law

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The same search, in every contract

Find the clause types

Assignment, change of control, termination, renewal and governing law, in every agreement.

Copy the wording and page

Quotations and page numbers go into a spreadsheet by hand.

Wonder about blanks

Does an empty result mean the clause is absent, or that the search missed it?

Maya, a junior associate, reviews 30 contracts before an acquisition. Her senior asks: “Where does the contract actually say that?”

These pain points are hypotheses, not measured findings. Maya is an illustrative persona.

From upload to a reviewed table

1
Upload
PDF or text contracts, and the categories to check.

2
Find
Candidate passages are found; only exact quotations are kept.

3
Review
Accept, remove or restore each finding, shown with its page.

4
Export
A CSV of the reviewed table, with every gap listed.

Each category in each contract ends in one of three visible states

Found | Not found, review required | Failed to process

If every finding in a category is removed, it returns to “Not found, review required”, so removing a false match never reads as a confirmed absence.

Review screen built in Streamlit; results stored in a local SQLite file.

Maya’s review, start to finish

1 **Upload once**
The 30 contracts go in together, with all five categories selected.

2 **Read the table**
Each contract and category is Found with its quotation and page, Not found, review required, or Failed to process.

3 **Check the wording**
She opens the contract at the quoted page, accepts correct findings and removes false matches.

4 **Search the gaps by hand**
Only the Not found cells need her own search, because an empty result never proves the clause is absent.

5 **Send the table**
She downloads the CSV and sends it, with the contracts, to her senior lawyer.

What her senior sees

One row per finding: contract, category, exact quotation, page and review state. A bracketed row marks every category nobody has confirmed.

“Where does the contract actually say that?” is answered by the page number.

He still decides what each clause means. The tool does not interpret.

Maya is an illustrative persona. Time saved is untested: the reviewer pilot was not run.

Two guarantees

100%

Verbatim quotations

A quotation is shown only if it is found word for word in the contract. Anything else is dropped.

100%

Visible gaps

Every contract and category without a verified finding is labeled for review. A missing result is never read as absence.

Both are enforced by the verify step and the status labels, not by the model.

Six steps from file to table

Step	What happens
Ingest	pypdf reads each page. Unreadable or scanned files become “Failed to process”.
Chunk	3,000-character windows with 300 characters of overlap.
Extract	One model call per chunk returns the category and the exact quotation. A keyword pass is kept only where the model found nothing.
Verify	Drop any quotation not found word for word. Where findings overlap, keep the model’s, then the tightest.
Store	Findings and category status go to a local SQLite file.
Export	Accept, remove or restore in the review screen, then download the CSV.

Model: Claude Sonnet 5.5, called through the facilitator’s API key.

Tested on CUAD, scored once

510
public contracts in CUAD

5
clause categories, mapped to CUAD labels

102
test contracts, held out until the freeze

1
test run, after the freeze tag

How a match is counted

A prediction is right when its overlap with a labeled passage, divided by their union, is at least 0.5. Each labeled passage matches at most one prediction.

Counts are pooled per category. The headline is the average of the five categories.

The other 408 contracts were used for development. Freeze tag: freeze-2026-11-08.

Headline result on 102 test contracts

Precision	Recall	F1
0.65	0.74	0.69
95% range 0.57 to 0.74	95% range 0.68 to 0.80	95% range 0.62 to 0.76

A regex-only baseline scores F1 0.42 on the same contracts. The extractor adds about 27 points.

One run, frozen code, Claude Sonnet 5.5. Ranges resample the 102 contracts.

Results vary widely by category

Category	Precision	Recall	F1	Right / predicted / labeled
Assignment	0.68	0.67	0.68	92 / 135 / 137
Change of Control	0.44	0.52	0.47	32 / 73 / 62
Termination for Convenience	0.69	0.71	0.70	29 / 42 / 41
Renewal	0.60	0.84	0.70	26 / 43 / 31
Governing Law	0.86	0.96	0.91	86 / 100 / 90
Average of five	0.65	0.74	0.69	

Lenient overlap rule: precision 0.69, recall 0.78, F1 0.73. No contract failed.

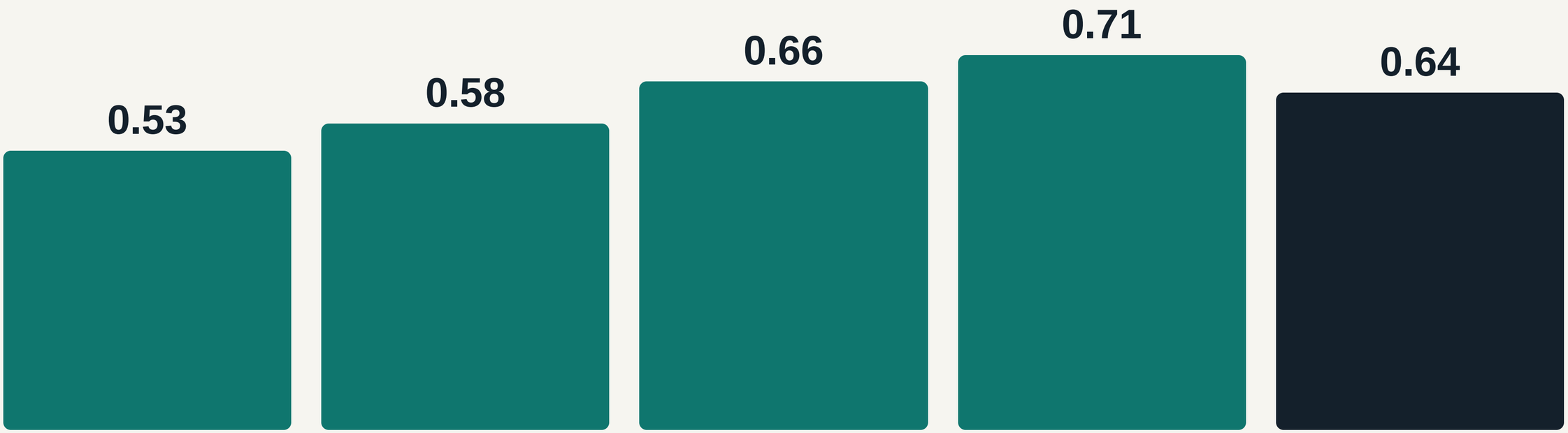
Results against our targets

Target	Result	Outcome
F1 at least 10 points above the regex baseline	0.69 against 0.42, about 27 points above	Met
Average recall at least 0.80	0.74 (lenient 0.78)	Not met
Average precision at least 0.70	0.65 (lenient 0.69); range reaches 0.74	Not met
30 largest contracts in 30 minutes or less	All 102 took 14 min 40 s; not timed separately	Met (inferred)
Reviewer pilot: median time 30% lower	Not run; no reviewers were available	Untested

Cost of the test run: $8.85 for 102 contracts, about $0.09 each.

2,096 model calls at $2 and $10 per million input and output tokens, 4 parallel workers.

How the extractor improved



Development contracts only; test split never used. About 7 points were specific to the tuning contracts.

We checked the answer key after the run

What was checked

Marks on 10 test contracts, compared with the CUAD labels. Three more contracts were hand-marked by one reviewer.

The 10 sets of marks came from an AI model, so this is not an independent human check.

10 items, all reviewed

Nine span differences: the same clause is found, but the highlighted passage differs.

One label disagreement: contract 09, Assignment (the hand mark was the wrong one).

Each item was judged against the signed definitions. We kept the CUAD labels as published (F1 0.691). As a check done after the freeze, removing the disputed spans moves F1 only to 0.695.

Reported scores use the CUAD labels as published. This check was made after the freeze.

Read these numbers with care

Precision is understated

CUAD leaves some real clauses unlabeled, and our Assignment definition is narrower than CUAD's.

Change of Control is weakest

F1 0.47. CUAD's own labeling is inconsistent for this category.

Answer-key check is not independent

The marks on 10 contracts came from an AI model; 3 contracts were hand-marked by one reviewer.

Time saved is untested

No reviewer pilot or practitioner interviews were done, so that target and the pain point are unproven.

A verification bug remains

Three correct quotes in 60 development contracts were dropped because page-break filler sits inside the sentence.

Public contracts only

The database stores full contract text, so real client contracts should not go to a hosted model.

Non-goals: legal interpretation, risk scoring, scanned PDFs, non-English contracts.

What we would change next

Nothing that affects the findings or scores was changed after the test run, so the frozen result stays clean. A definition change needs approval and a new version tag.

Assignment

Count sentences that make a prohibited assignment void or a breach. Development precision 0.71, recall 0.68.

Termination for convenience

Count a without-cause right to end part of the deal, such as a license or a fund. Precision 0.66, recall 0.78.

Change of control

Little room expected, because CUAD’s labeling is inconsistent here. Precision 0.35, recall 0.53.

Verification fix

Ignore page-break filler when matching quotes. Removed findings are now kept in the CSV (done Nov 10, scores unaffected).

These errors come from 100 development contracts, not the test split.